PRODUCT CASE STUDY

Bulk User Information Update

Enabling administrators to modify attributes across dozens to hundreds of employees at once — with validation, approval workflows, scheduling, and downstream propagation.

WORKING PROTOTYPE | 20+ EDGE CASES | DATA PROPAGATION

QUESTION 1 — APPROACH

Three Approaches Evaluated

OPTION A

Spreadsheet-Only

Download CSV → edit offline → re-upload.

✓ PROS

- Handles per-user values
- Familiar Excel workflow

✕ CONS

- Error-prone
- No real-time validation
- Bad for uniform changes

OPTION B

Filter + Rules Only

Select via filters → apply uniform operations.

✓ PROS

- Fast for uniform changes
- In-app validation

✕ CONS

- Can't handle per-user values
- Doesn't scale for mixed changes

✓ CHOSEN

Hybrid: Both Methods

User picks the best tool for their use case within one flow.

✓ PROS

- Best of both worlds
- All use cases covered
- Flexible for any batch size

✕ CONS

- More complex to build

Why Hybrid: One flow, two ways to edit. **Filters** = same change for many people (e.g. 5% raise for all L3s). **CSV** = different value per person (e.g. each employee's new comp). Pick the one that fits; the rest stays simple.

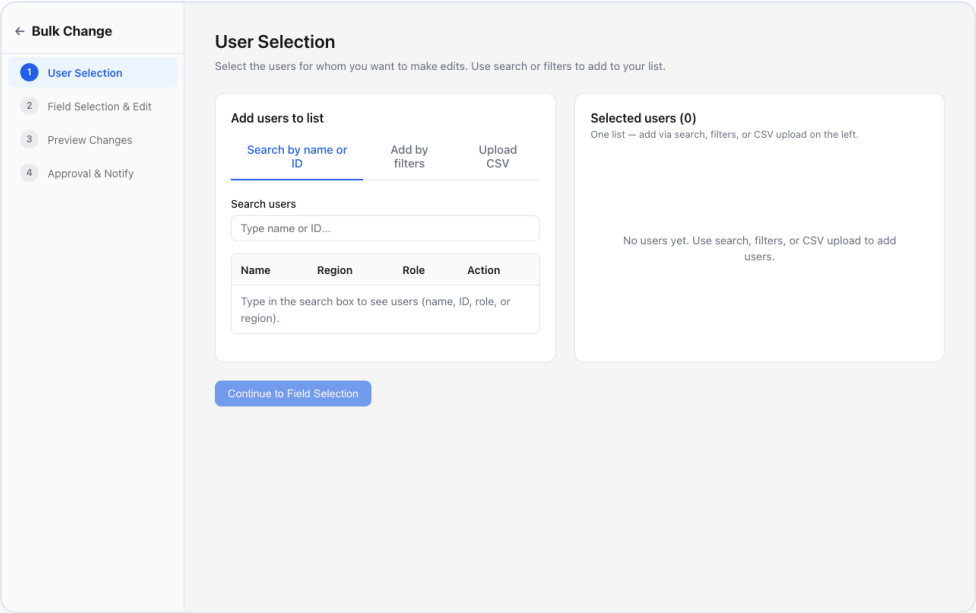
QUESTION 2 — PRODUCT FLOW

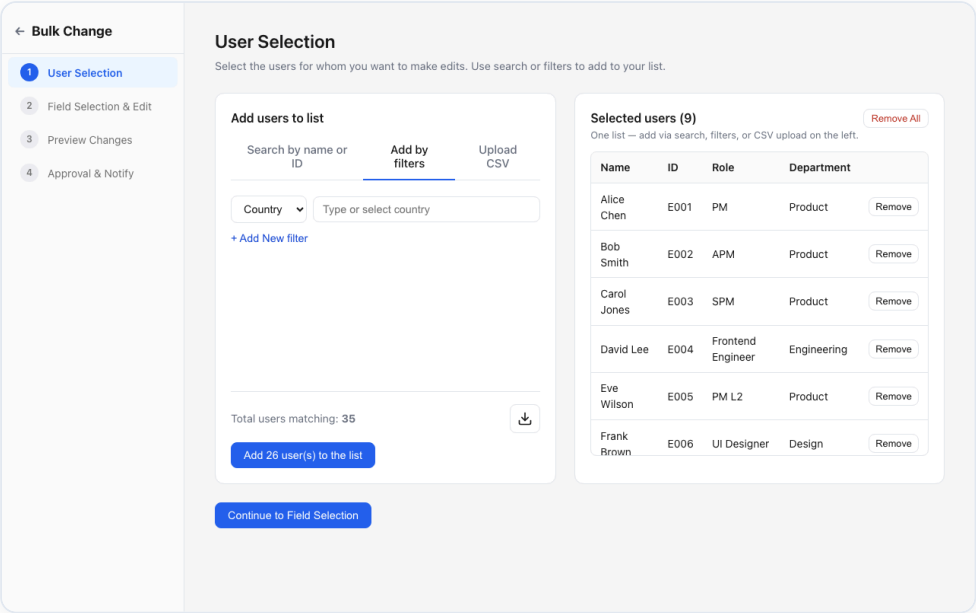
Flow Walkthrough

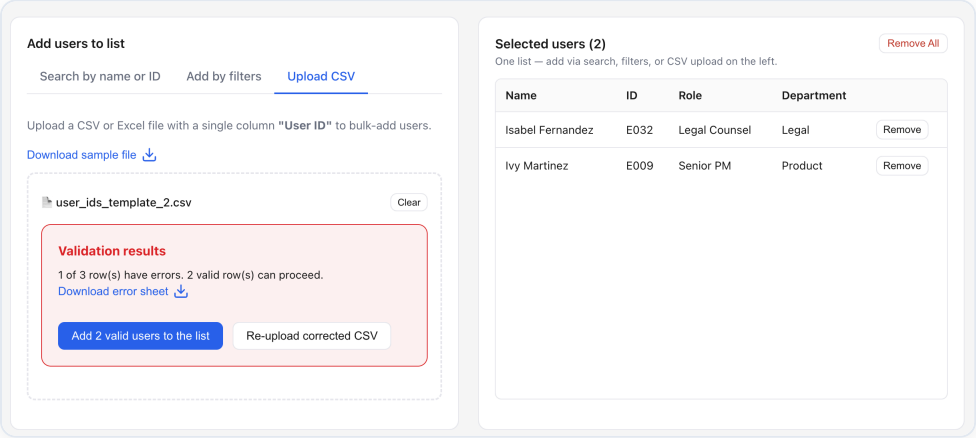
A guided 4-step wizard: Select Users → Define Changes → Preview & Submit → Status

📋 Edit History → 👥 Select Users → ✏️ Define Changes → 👁️ Preview → ✅ Status

Step 1 Select Users





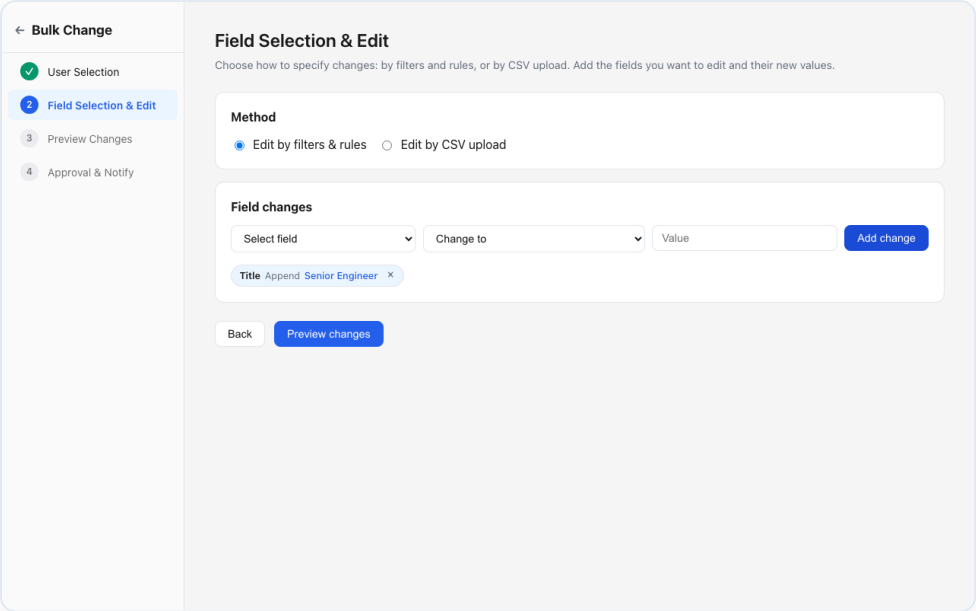


Edge Cases

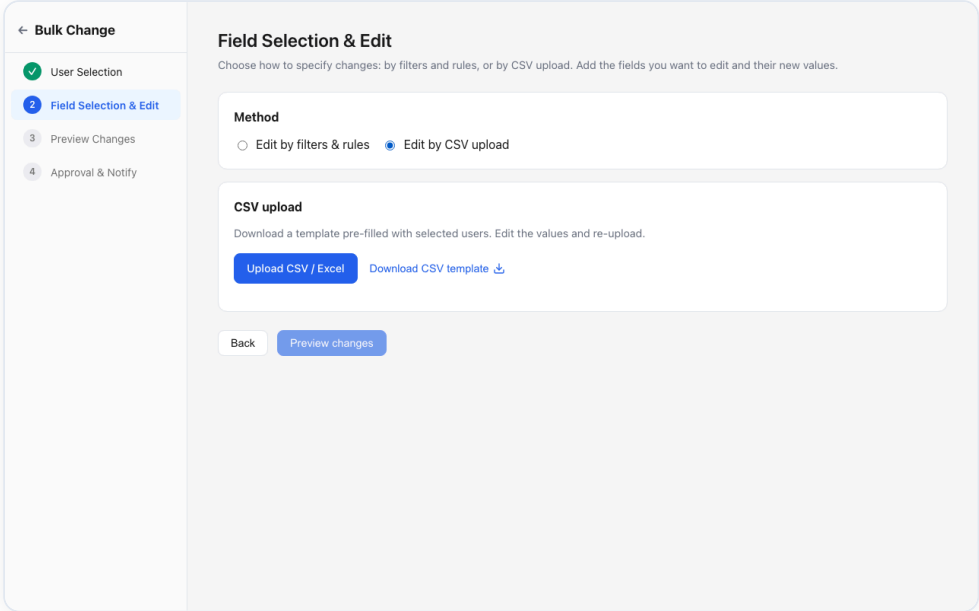
- ✓ **Duplicate prevention** — "Already added" inline
- ✓ **Smart CTA** — disables when all filtered users already selected
- ✓ **Filters persist** after applying — not cleared
- ✓ **Remove All** — one-click option to clear entire selected users list

Step 2 Define Changes

Method A: Filter + Rules — apply uniform operations.



Method B: CSV Upload — per-user values.



12 fields · 10 operations · Sensitive fields flagged (Compensation, Currency, Status)

Edge Cases

- ⚠ **Currency conflict** — blocks absolute ops for mixed currencies
- ⚠ **Duplicate field** — warns when same field added again
- ! **File size** — 5 MB max · format: .csv/.xls/.xlsx only
- ✓ **Partial CSV errors** — continue with valid rows, exclude bad ones
- ✓ **Auto-strips "Errors" column** on corrected re-upload

CSV WORKFLOW

Template → Upload → Validate → Error Handling

📥 Template Download

Pre-filled with selected users. Headers: User ID, Name, Compensation, Title, Department, Location.

❌ Error Sheet

"Errors" column prepended · Red/green color coding · Download as .xls

✅ Final Change Sheet

Only rows with changes · Modified cells highlighted yellow · Available from Edit History

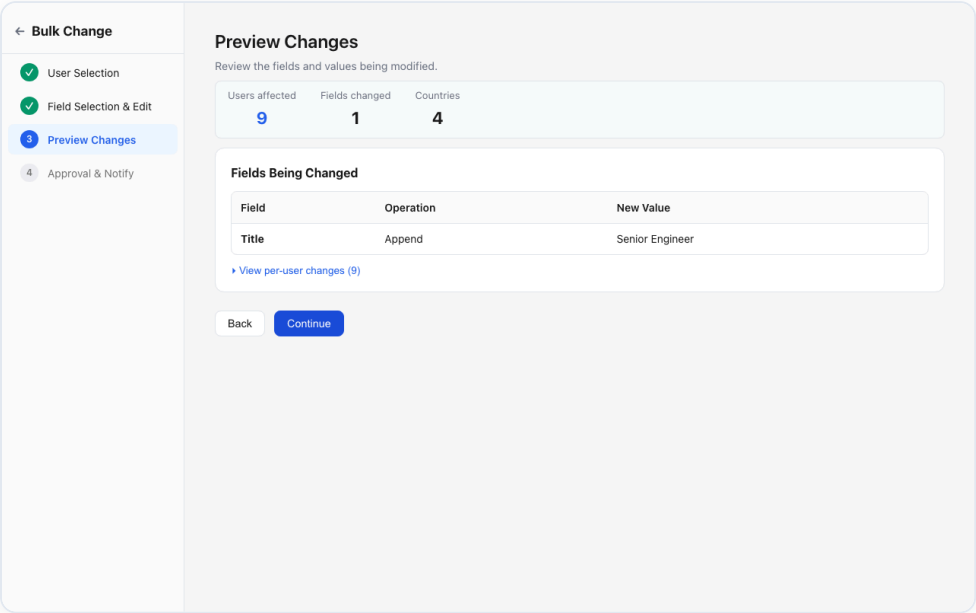
Validation Rules

- **Headers:** must match template exactly
- **User ID:** non-empty, must exist in selected users
- **Compensation:** valid positive number
- **Required fields:** Name, Title, Dept, Location
- **File size:** max 5 MB
- **Format:** .csv, .xls, .xlsx only

Recovery Options

- "Continue with N valid rows" — excludes errors
- "Download error sheet" — for offline correction
- "Re-upload corrected" — auto-strips Errors column

Step 3a Preview — Review Changes



What's Shown

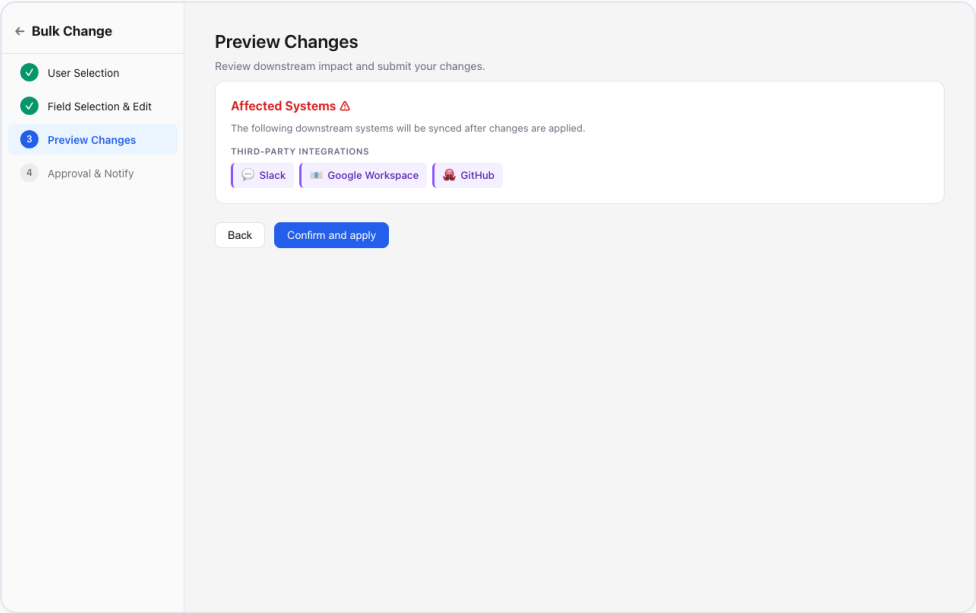
- **Impact stats:** users affected, fields changed, countries, excluded
- **Fields table:** each field, operation, value · "Sensitive" badge on critical fields
- **Per-user preview:** expandable old → new values for each user

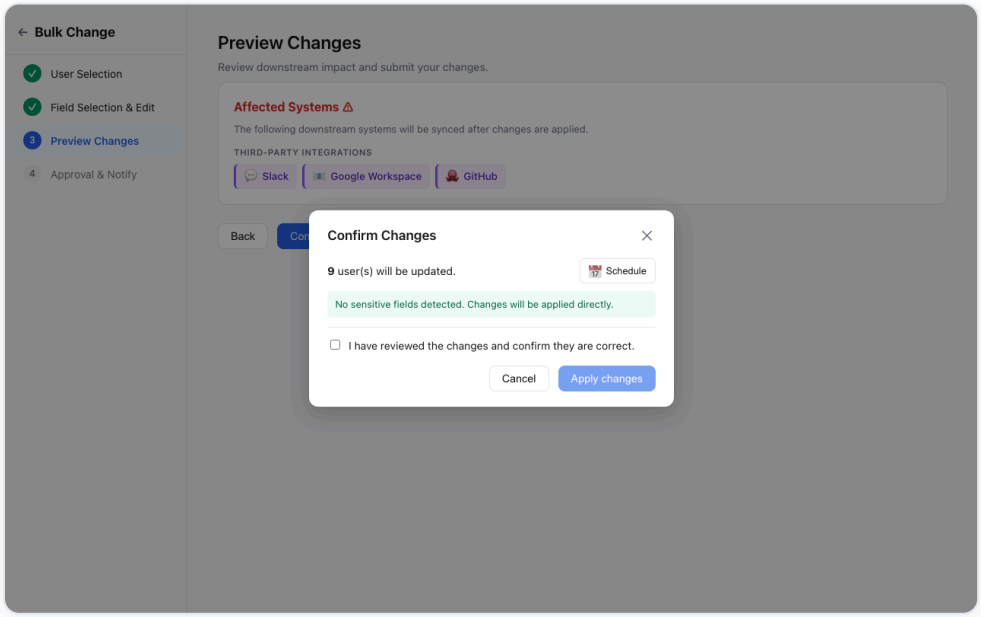
Edge Cases

! **Zero edits** — redirects back to Step 2

! **Zero users after exclusions** — submit disabled

Step 3b Preview — Impact & Submit





Affected Systems ⚠

Field → system mapping shows exactly which downstream systems will sync.

💰 Payroll | 🏥 Benefits | 💻 Devices | 💬 Slack | 📧 Google | 🐙 GitHub

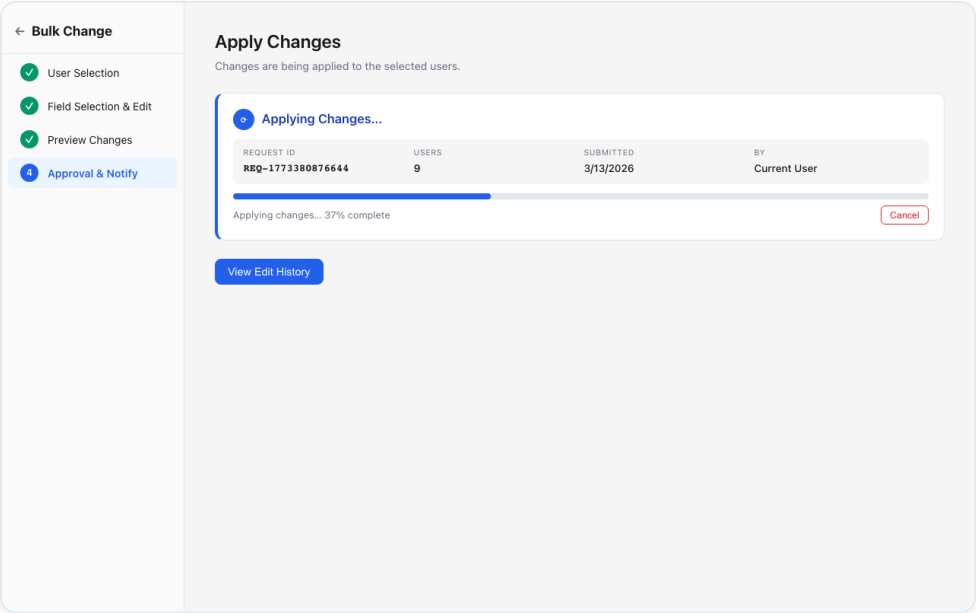
Confirmation Modal

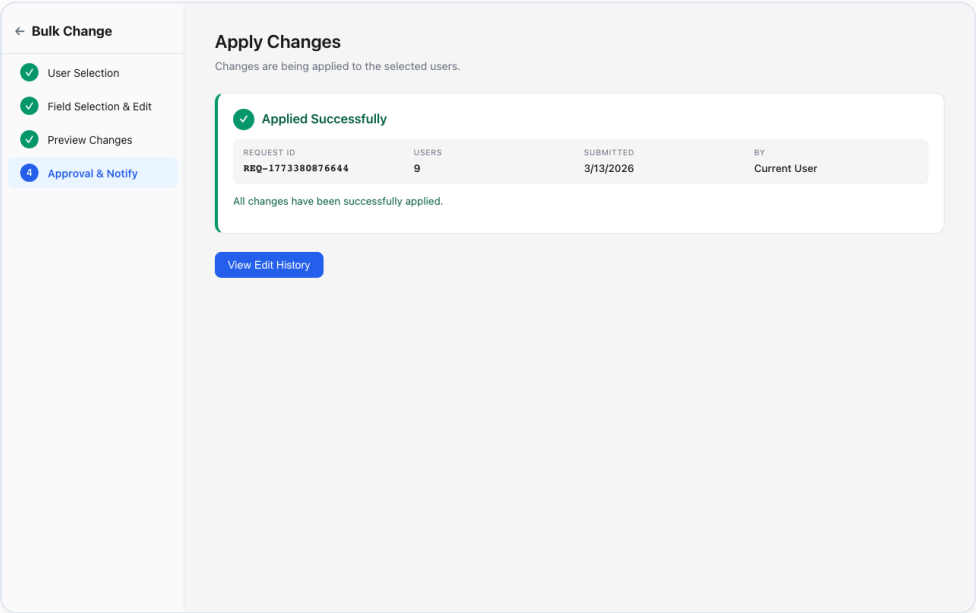
- User count + sensitive field warning
- 📅 Schedule button — pick future date
- Checkbox confirmation required
- Dynamic CTA: "Submit for approval" / "Apply" / "Schedule"

Scheduling

Scheduling never bypasses approval. If sensitive fields → approval required regardless of date.

Step 4 Approval & Application





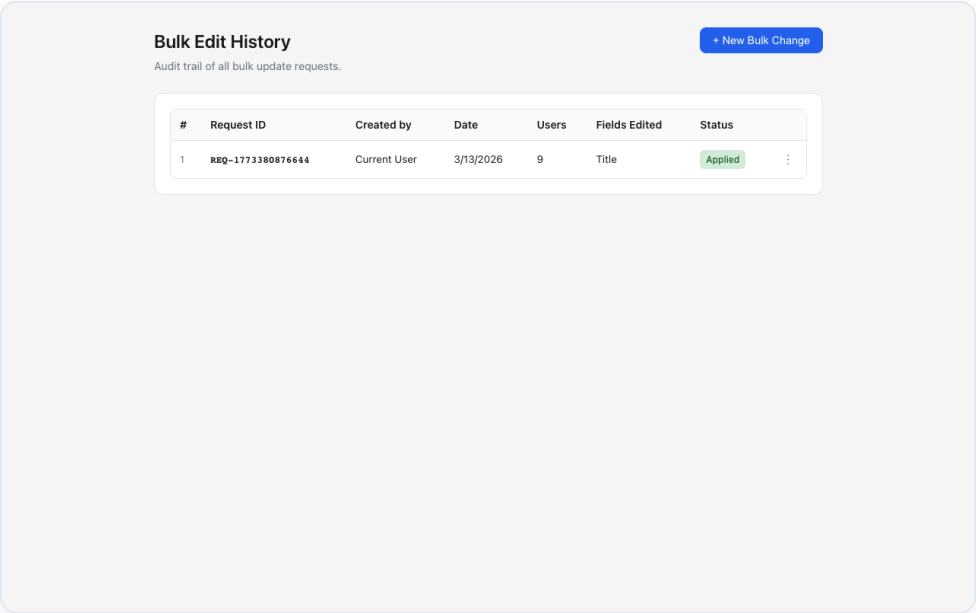
Four Paths

SENSITIVE?	SCHEDULED?	RESULT
No	No	APPLY IMMEDIATELY
No	Yes	SCHEDULED
Yes	No	PENDING APPROVAL
Yes	Yes	SCHED + PENDING

Edge Cases

- ⚠ **Partial failure** — ~15% fail. Retry or continue without.
- ✓ **Cancel mid-processing** — stops immediately, no changes applied
- ✓ **Failed count resets** on retry
- ! **Reversal blocked** for sensitive fields (comp, currency, status)

Home Edit History — Audit Trail



Features

- Full table: Request ID, Created By, Date, Users, Fields, Status
- 📩 **Download change sheet** — styled .xls, only changed rows, cells highlighted
- ↩️ **Reverse changes** — blocked for sensitive fields
- **+ New Bulk Change** — resets and starts fresh

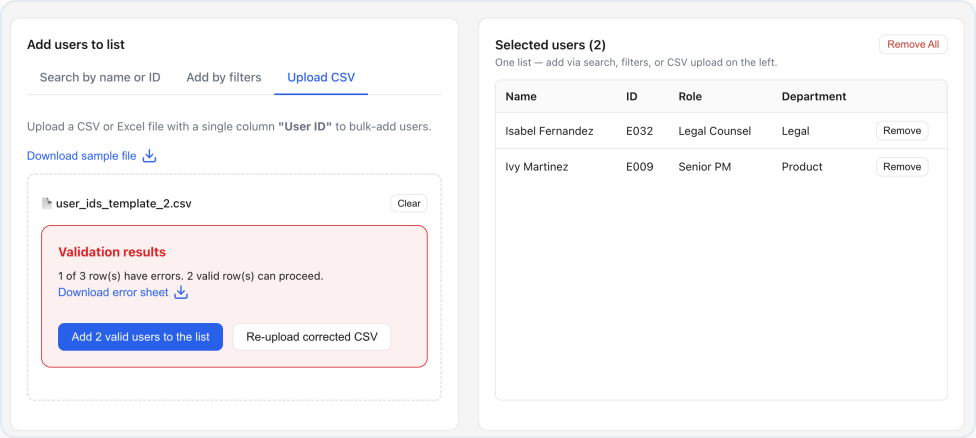
Status Badges

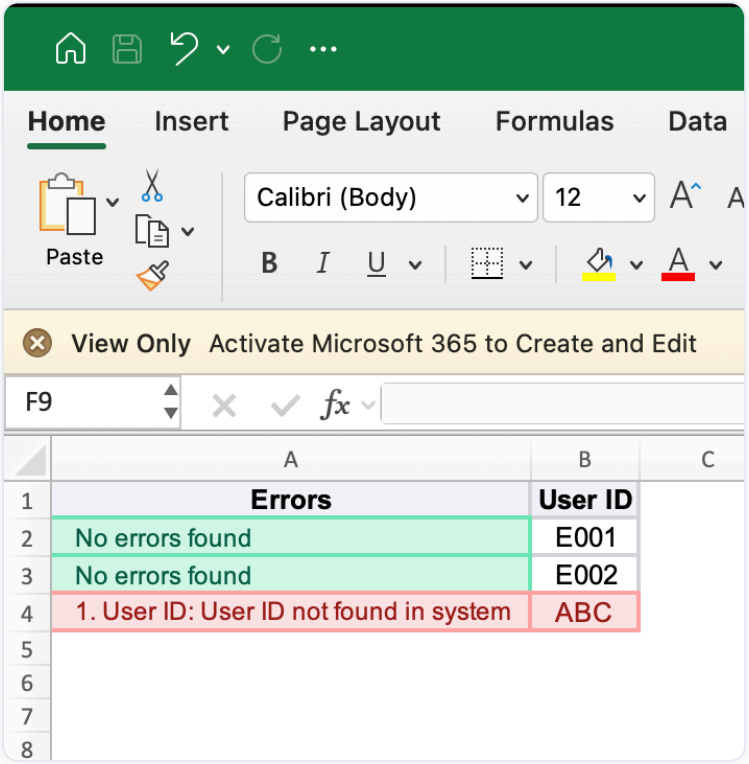
APPLIED PARTIAL PENDING SCHEDULED REJECTED CANCELLED

SCREENSHOTS — ERROR HANDLING & CSV

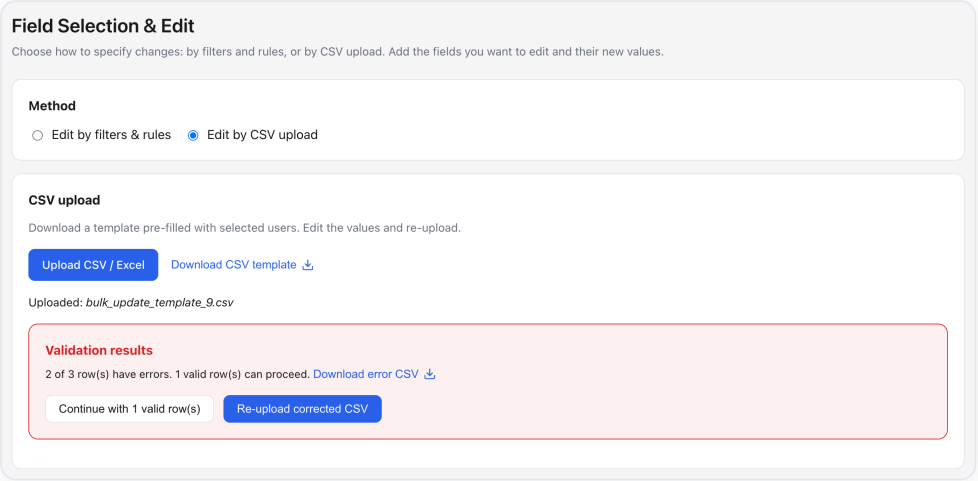
User & Field Selection CSV Validation

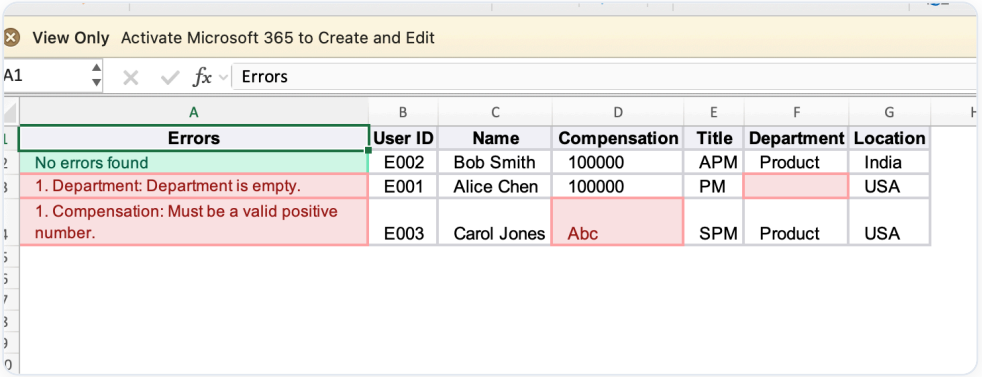
User upload (Step 1)





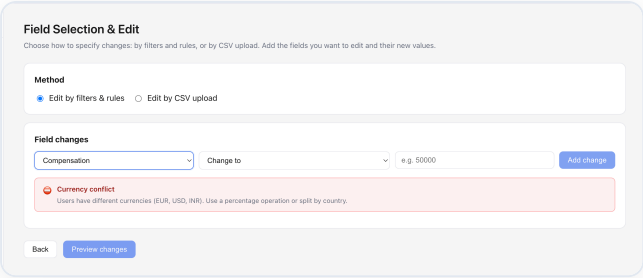
Field Selection & Edit CSV

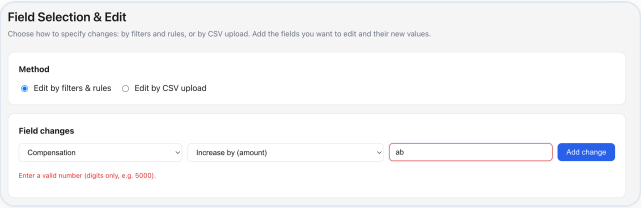


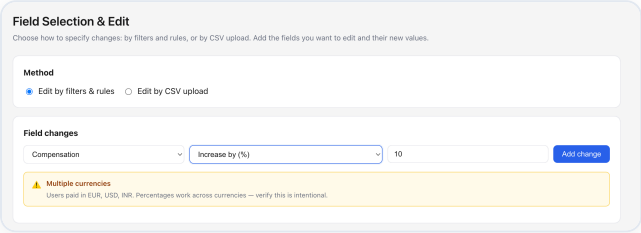


bulk_update_template (9)

User ID	Name	Compensation	Title	Department	Location
E002	Bob Smith	100000	APM	Product	India
E001	Alice Chen	100000	PM	Product	USA
E003	Carol Jones	100000	SPM	Product	USA



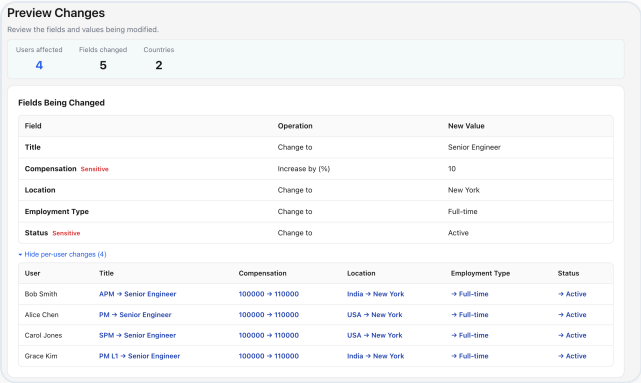




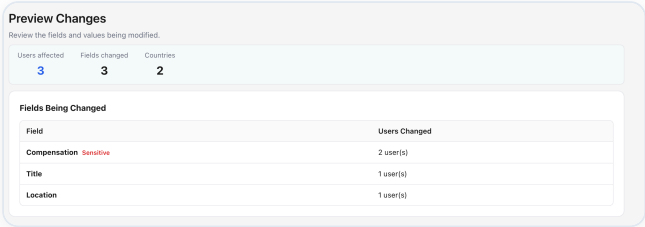
Download sample template, validation failures, error sheet (Errors column + red cells), and inline errors/nudges.

SCREENSHOTS — PREVIEW & APPROVAL

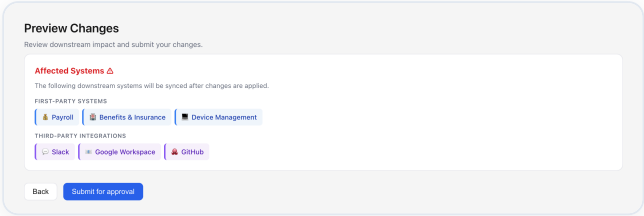
Preview Changes & Approval Flows



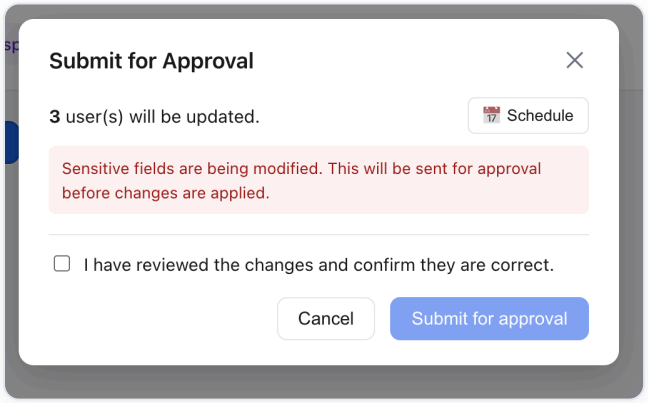
Preview — Rules



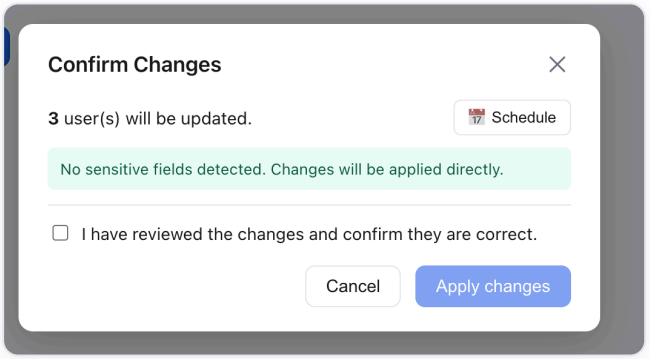
Preview — CSV



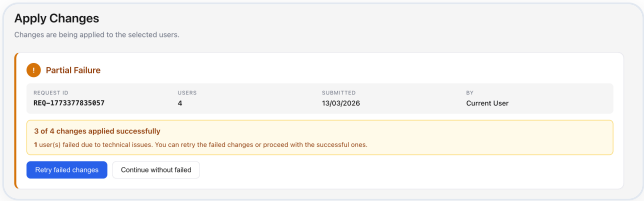
Affected Systems



Submit for approval



Approval not needed



Partial approval

SCREENSHOTS — STATUS & ACTIONS

Scheduling, Edit History & Reverse

Scheduling

Scheduled Change

Your changes have been scheduled for a future date.

Scheduled

REQUEST ID	USERS	SUBMITTED	BY
REQ-1773378068948	3	13/03/2026	Current User

Scheduled for Monday, March 16, 2026

Changes will be automatically applied on the scheduled date. You will be notified once processing begins.

Scheduled Change

Your changes are scheduled and awaiting approval.

Scheduled – Pending Approval

REQUEST ID	USERS	SUBMITTED	BY
REQ-1773378207844	1	13/03/2026	Current User

Scheduled for Monday, March 16, 2026

Sensitive fields are being modified. Changes will be applied on the scheduled date only after approval is granted.

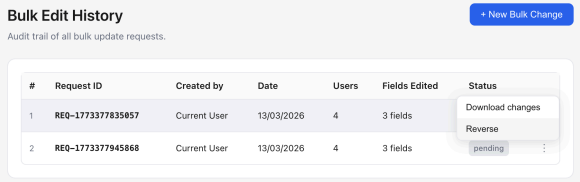
Edit History & Reverse

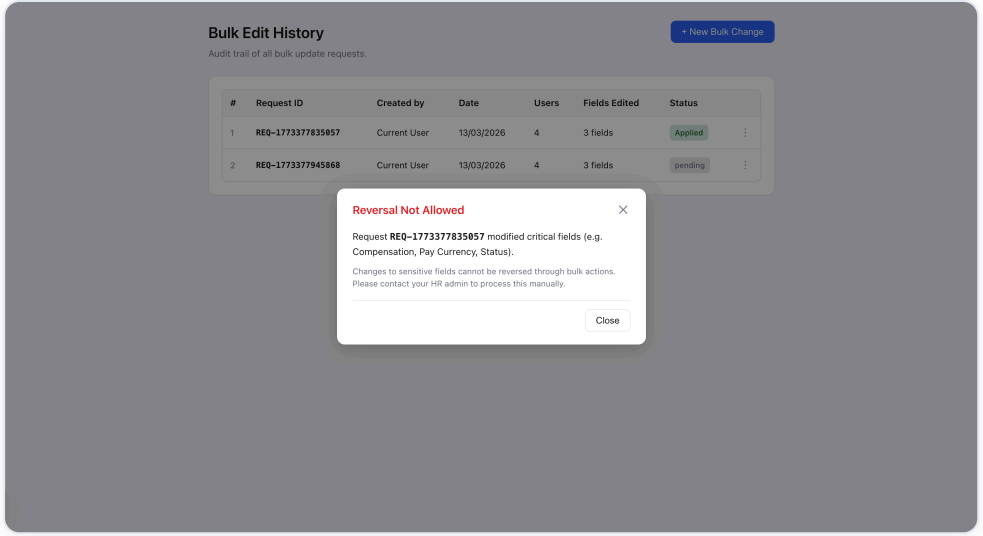
Bulk Edit History

+ New Bulk Change

Audit trail of all bulk update requests.

#	Request ID	Created by	Date	Users	Fields Edited	Status
1	REQ-1773377835057	Current User	13/03/2026	4	3 fields	Applied
2	REQ-1773377945868	Current User	13/03/2026	4	3 fields	pending
3	REQ-1773378034842	Current User	13/03/2026	3	Title	Applied
4	REQ-1773378068948	Current User	13/03/2026	3	Department	Scheduled Mar 16, 2026
5	REQ-1773378207844	Current User	13/03/2026	1	Compensation	Pending Approval Mar 16, 2026




Bulk Edit History
Audit trail of all bulk update requests.
+ New Bulk Change
# Request ID Created by Date Users Fields Edited Status
1 REQ-1773377835057 Current User 13/03/2026 4 3 fields Applied
2 REQ-1773377945868 Current User 13/03/2026 4 3 fields pending
Reversal Not Allowed
Request REQ-1773377835057 modified critical fields (e.g. Compensation, Pay Currency, Status).
Changes to sensitive fields cannot be reversed through bulk actions. Please contact your HR admin to process this manually.
Close

QUESTION 3

Data Propagation Challenges & Solutions

Challenges

1. Atomicity

Comp change must hit Payroll AND Benefits together. Partial sync = compliance risk.

2. Rate Limiting

500 users × 3 integrations = 1,500+ API calls. Third-party limits cause sync delays.

3. Ordering

Status changes: Payroll before Device Management. Bulk amplifies dependency complexity.

4. API Outages

GitHub down shouldn't block the entire change. But partial propagation creates inconsistency.

Solutions

🔄 Async Job Queue + Retry

Each downstream sync = independent job. Exponential backoff. Decouples core change from propagation.

🔀 Dependency Graph (DAG)

Process in topological order. Parent fails → child syncs held.

🛡️ Idempotent Operations

Every sync safe to retry. Change-request-ID for deduplication.

📅 Scheduling as Buffer

Batch during off-peak hours. Reduces rate-limit pressure. Lands before payroll cut-offs.

QUESTION 3 — CONTINUED

Field → System Impact Matrix

This mapping drives the "Affected Systems" preview shown to admins before submit.

FIELD	FIRST-PARTY	THIRD-PARTY	RISK
Compensation	Payroll, Benefits	—	HIGH
Status	Payroll, Benefits, Devices	Slack, Google, GitHub	HIGH
Title	—	Slack, Google, GitHub	LOW
Department	Cost Center	Slack, Google	MED
Location	Payroll, Benefits, Devices	—	MED
Manager / Team	—	Slack, Google	LOW

Why this matters: Admins see exactly which systems are affected before submitting. They can schedule for off-hours if many critical systems are involved.

SUMMARY

Key Takeaways

🔀 Hybrid Approach
Filter + Rules & CSV. One flow, two paths.

🛡️ Smart Approval
Only sensitive fields need approval. Scheduling never bypasses it.

⚠️ 20+ Edge Cases
Currency conflicts, partial failures, file validation, cancellation.

🔗 Propagation Aware
Field → system mapping. Async queues, retry, dependency ordering.

📅 Scheduling
Future dates. Off-peak processing. Approval + schedule independent.

📋 Full Audit Trail
Every request tracked. Change sheets. Reversals with guardrails.

Working prototype available for live demo →